

Conversational Business Intelligence for FMCG Category Management

AI-Powered SQL Assistant for Sales & Inventory Analytics

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The Business Challenge

Daily Data Requests

Brand Managers and Category Directors constantly request promotional lifts, inventory stockouts, and regional comparisons.

Bottlenecks & Delays

Data teams face excessive loads answering simple questions manually, causing decision latency of days.

⚡ **Self-Service Focus:**
Transition non-technical category managers to conversational discovery.

⚡ **Standard Aggregations:**
Automate weekly transactional queries.

⚡ **Proactive Warnings:** Auto-detect inventory stockout indicators.

Solution Architecture

Natural Language to SQL

System prompts inject the SQLite schema description into Gemini to translate natural queries into valid SELECT queries dynamically.

SQLite Execution Engine

Express server runs the validated query, compiles a spreadsheet grid, and auto-calculates horizontal bar chart values.

Dynamic Orchestration Flow

1. User Question in UI
2. LLM SQLite Query Translation
3. Database Execution & Data Fetch
4. LLM Narrative Summary Formulation
5. Tabular & Visual Chart Displays

Simulated FMCG Data Sourcing

- ⚡ **Scale & granularity:** Models 15 products across 5 categories, 24 stores, 4 regions, and 16 weeks of weekly sales and inventory history (7,680 records).
- ⚡ **Coherent Stock Logic:** Models structural inventory dependencies:
`closing_stock = opening + received - sold.`

Promotion Multipliers

Includes high-uplift multipliers (4x-5x on BOGO, 3x-4x on Price Cuts) to verify SQL query reasoning correctness.

Stockout Modeling

Forces delivery delay flags (units received = 0) to verify stockout reasoning and inventory warning outputs.

Conversational UI & Explorer

Interactive AI Chat

Enables prompt requests with simple suggestions. Returns narrative analysis in markdown formatting.

Auto-Charting Engine

UI automatically identifies numeric and label columns in query responses and renders horizontal CSS charts without extra package weight.

Trust & Debug Toggles

Allows users to toggle open the exact executed SQL query and inspect the full raw database grid, promoting transparency.

Risk Control & Future Roadmap

Current System Failure Points

- ⚡ Breaks on vague requests without numeric parameters.
- ⚡ File-based SQLite locks under concurrent write queries.

Version 2.0 Enhancements

- ⚡ **Semantic Cache:** Map similar prompts to SQL to reduce LLM costs.
- ⚡ **Table Export:** Export tables straight to Excel files.
- ⚡ **Multi-Agent Verification:** Add verification loop for SQL aggregation safety.